

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 04

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Recap of Previous Lecture



- ✓ Topic Cartesian product
- ✓ Topic Relation
- ✓ Topic Diagonal relation, Reflexive and irreflexive relation
 $2^{\frac{n(n+1)}{2}}$ $2^{(n^2-n)}$ $2^{(n^2-n)}$
- ✓ Topic Symmetric, anti-symmetric and asymmetric relation
 2^n $3^{\frac{n(n-1)}{2}}$ $3^{\frac{n(n-1)}{2}}$
- ✓ Topic Transitive relation

Topics to be Covered



✓ Topic

Transitive relation

✓ Topic

Reflexive closure

✓ Topic

Symmetric closure

✓ Topic

Transitive closure

Topic

Complement of a relation and Inverse of a relation



Topic : Types of Relations

All these relations are defined on same set, i.e., from a set A to set A itself

- (1) Diagonal Relation (Identity Relation)
- (2) Reflexive Relation
- (3) Irreflexive Relation
- (4) Symmetric Relation
- (5) Anti-symmetric Relation
- (6) Asymmetric Relation
- (7) Transitive Relation

- (8) Complement of a relation
- (9) Inverse of a relation
- (10) Composite of two relations

Today's Topic



Topic : Transitive Relation

A relation R on set A is called transitive relation if and only if

if $(a^R b \text{ and } b^R c)$ then $a^R c, \forall a, b, c \in A$

i.e. if $((a, b) \in R \text{ and } (b, c) \in R)$ then $(a, c) \in R, \forall a, b, c \in A$

H.W

eg:- Let $A = \{1, 2, 3\}$

- ① $R_1 = \{ \}$ Empty relation is the smallest transitive relⁿ
- ② $R_2 = A \times A$ ' $A \times A$ ' is the largest transitive relⁿ on set A
- ③ $R_3 = \{ (1,1), (2,2) \}$ It is a transitive relⁿ

Note: Let $(a,a) \in R$

∴ $(a,a) \in R$ and $(a,a) \in R$. then

because of transitivity (a,a) must
be present in the relation

and (a,a) is already present

Note: Let $(a,a) \in R$ and $(a,b) \in R$



then by transitivity (a,b) must be present in $\mathcal{R}^n$
and (a,b) is already present in $\mathcal{R}^n$

Note:

Let $(b,a) \in R$ and $(a,a) \in R$

then by transitivity (b,a) must be present in R^n
and (b,a) is already present in R^n

H.W

let $A = \{1, 2, 3\}$

(diagonal) $\circ \circ$ No problem w.r.t $(1, 1)$

④ $R_4 = \{ (1, 2), (2, 1), (1, 1) \}$

Check whether R_4 is transitive or not

Ends with '2'

$\circ \circ$ We need to check with order pairs starting with '2' {i.e., $(2, 1)$ }
 $(1, 2) \& (2, 1) \therefore$ By transitivity $(1, 1)$ must be present

$\hookrightarrow$ Already present $\therefore$ No problem

Ends with '1'

$\circ \circ$ We need to compare with order pairs that starts with '1' (i.e., $(1, 1) \& (1, 2)$)

$(2, 1) \& (1, 1) \circ \circ$ By transitivity $(2, 1)$

$\hookrightarrow$ it is already present

$(2, 1) \& (1, 2) \circ \circ$ By transitivity $(2, 2)$

$\hookrightarrow$ But $(2, 2) \notin R_4$ Hence R_4 is not transitive

H.W.
eg.

$$A = \{1, 2, 3\}$$

$$R_5 = \{(1, 2), (1, 3)\}$$

transitive or not?

Ends with '2'

But no order pair starts with '2'

∴ No problem w.r.t. (1, 2)

it Ends with '3'

And no order pair starts with '3'

∴ No problem w.r.t. (1, 3)

No order pair dis-satisfy the condⁿ of transitivity

∴ R_5 is a transitive Relⁿ

H.W.

eg: $A = \{1, 2, 3\}$

$R_6 = \{ (2, 3), (1, 2) \}$

transitive or not?

Ends with '2'

$(1, 2) \neq (2, 3)$

$\therefore$ By transitivity $(1, 3)$

but $(1, 3) \notin R_6$

$\therefore R_6$ is not transitive

H.W eg 1: Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

transitive or not?

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

$\uparrow$
 $\therefore (3,1) \neq (1,2) \therefore (3,2)$

but $(3,2) \notin R$
 $\therefore$ Not transitive

H.W.

eg 2: Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

transitive or not?

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\} \quad \text{it is a transitive relation}$$

eg. Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$$

transitive or not?

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$$

↑
'3' & (3,2) ∴ (1,2)

↓
but $(1,2) \notin R$

∴ Not transitive

eg. Let $A = \{a, b, c, d\}$

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

transitive or not?

$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

'd' $\notin (d,c) \therefore (a,c)$

↓
but $(a,c) \notin R$

$\therefore$ Not transitive



Topic : Reflexive Closure

- Let R is any relation on set A .
- Reflexive closure of relation R is the smallest reflexive relation on set A that contains relation R

eg. let $A = \{1, 2, 3\}$
 $R = \{(1, 1), (1, 2), (2, 1), (3, 2), (3, 3)\}$

Reflexive
Closure of $R = \{(1, 1), (1, 2), (2, 1), (3, 2), (3, 3), (2, 2)\}$

Note $\rightarrow$ ① If given relation R is already a reflexive relation on set A , then reflexive closure of relation R will be relation R itself.

* ② Reflexive closure of $R = R \cup \Delta_A$
Where R is any relation on set A



Topic : Symmetric Closure

- Let R is any relation on set A ,
Symmetric closure of relation R is the smallest symmetric relation on set A that contains R

eg. Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$$

Symmetric Closure of $R = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2), (1,3), (2,3)\}$

Note: { ① If R is symmetric relation on set A ,
then symmetric closure of relation R will be
relation R itself



Topic : Transitive Closure

- Let R is any relation on set A ,
Transitive closure of relation R is the
Smallest transitive relation on set A that
Contains relation R

eg 1: Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

Find transitive closure of R

1st iteration $\rightarrow R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

$$R^* = \{(1,1), (1,2), (1,3), (2,2), (3,1) \mid (3,2), (3,3)\}$$

2nd iteration $\rightarrow R^{**} = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$

We will keep iterating until we reach to a situation when no new order pair is added in the last iteration, and in that case the

result of the last iteration is the transitive closure of R .

No new order pair needs to be added to satisfy transitivity
 $\therefore R^{**}$ is transitive closure of R

eg 2: Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

Find transitive closure of R

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

1st iteration

$$R^* = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

No new order pair
needs to be added
to satisfy transitivity

∴ R^* is transitive
closure of R

and $R^* = R$

∴ R is a transitive Relⁿ

eg. Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$$

Find transitive closure of R .

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$$

1st iteration $\rightarrow R^* = \{(1,1), (1,3), (2,2), (3,1), (3,2) \mid (1,2), (3,3)\}$

2nd iteration $\rightarrow R^{**} = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$

No new
order pair
needs to
be added

$\therefore R^{**}$ is transitive closure of R

eg. Let $A = \{a, b, c, d\}$

$$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$$

Find transitive closure of R

$$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$$

$$\checkmark R^* = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c) \mid (a,c), (b,d), (c,c), (d,a), (d,d)\}$$

$$R^{**} = \{(a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d) \mid (a,a)\}$$

$$R^{***} = \{(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)\}$$

$\therefore R^{***}$ is transitive closure of R .

No new
order
pair
is
added

{ Warshall's algo }

Warshall's algo can be used to find the transitive closure of given relation.

eg. Let $A = \{a, b, c, d\}$

✓ $R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$

Find transitive closure of R

Matrix
w.r.t. relation R

(x,y) →

	a	b	c	d
a	0	0	0	1
b	1	0	1	0
c	1	0	0	1
d	0	0	1	0

← (a,d)

	I st	II nd	III rd	IV th
Column	$\{b, c\}$	$\{ \}$	$\{b, d\}$	$\{a, c\}$
Row	$\{d\}$	$\{a, c\}$	$\{a, d\}$	$\{c\}$
Column × Row =	(b,d) (c,d)	$\emptyset$	(b,a) (b,d) (d,a) (d,d)	(a,c) (c,c)

$R^* = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c),$
 $(b,d), (d,a), (d,d), (a,c), (c,c)\}$

$$R^* = \{ (a,d), (b,a), (b,c), (c,a), (c,d), (d,c) \\ (b,d), (d,a), (d,d), (a,c), (c,c) \}$$

Matrix
w.r.t. relation $R^* =$

(x,y)	a	b	c	d
a	0	0	1	1
b	1	0	1	1
c	1	0	1	1
d	1	0	1	1

Column $\times$ Row =

$$R^{**} = \{ (a,d), (b,a), (b,c), (c,a), (c,d), (d,c) \\ (b,d), (d,a), (d,d), (a,c), (c,c) \\ (a,a) \}$$

	I st	II nd	III rd	IV th
Column	{b,c,d}	$\emptyset$	{a,b,c,d}	{a,b,c,d}
Row	{c,d}	{a,c,d}	{a,c,d}	{a,c,d}
	(b,c) (b,d) (c,c) (c,d) (d,c) (d,d)	$\emptyset$	(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a), (c,c) (c,d), (d,a) (d,c), (d,d)	(a,a), (a,c) (a,d), (b,a) (b,c), (b,d) (c,a), (c,c) (c,d), (d,a) (d,c), (d,d)

$$R^{**} = \{ (a,d), (b,a), (b,c), (c,a), (c,d), (d,c), (b,d), (d,a), (d,d), (a,c), (c,c), (a,a) \}$$

Matrix w.r.t. relation R^{**} =

	a	b	c	d
a	1	0	1	1
b	1	0	1	1
c	1	0	1	1
d	1	0	1	1

(x,y) → (a,d)

$$R^{***} = \{ (a,d), (b,a), (b,c), (c,a), (c,d), (d,c), (b,d), (d,a), (d,d), (a,c), (c,c), (a,a) \}$$

	I st	II nd	III rd	IV th
Column	{a, b, c, d}	∅	{a, b, c, d}	{a, b, c, d}
Row	{a, c, d}	{a, c, d}	{a, c, d}	{a, c, d}
Column × Row =	(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)	∅	(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)	(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)

No new Order Pair is added

∴ R^{***} is transitive closure of R .



Topic : Complement of a relation



Complement is
always defined w.r.t.
Universal set

→ Let 'R' is a relation from
Set A to Set B,
then complement of relation R is
denoted by R^c and it is
defined as.

$$R^c = A \times B - R$$

$$A^c = U - A$$

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

eg. $A = \{1, 2\}$ $B = \{a, b, c\}$
And R is a relation from Set A to
Set B,

$$R = \{(1, a), (2, b), (2, c)\}$$

then $R^c = \{(1, b), (1, c), (2, a)\}$



Topic : Inverse of a relation

- Let R is a relation from set A to set B .
Inverse relation of relation R can be denoted by R^{-1} ,
and R^{-1} will be a relation from set B to set A
which is defined by

$$R^{-1} = \{ (b, a) \mid (a, b) \in R \}$$

from set A from set B

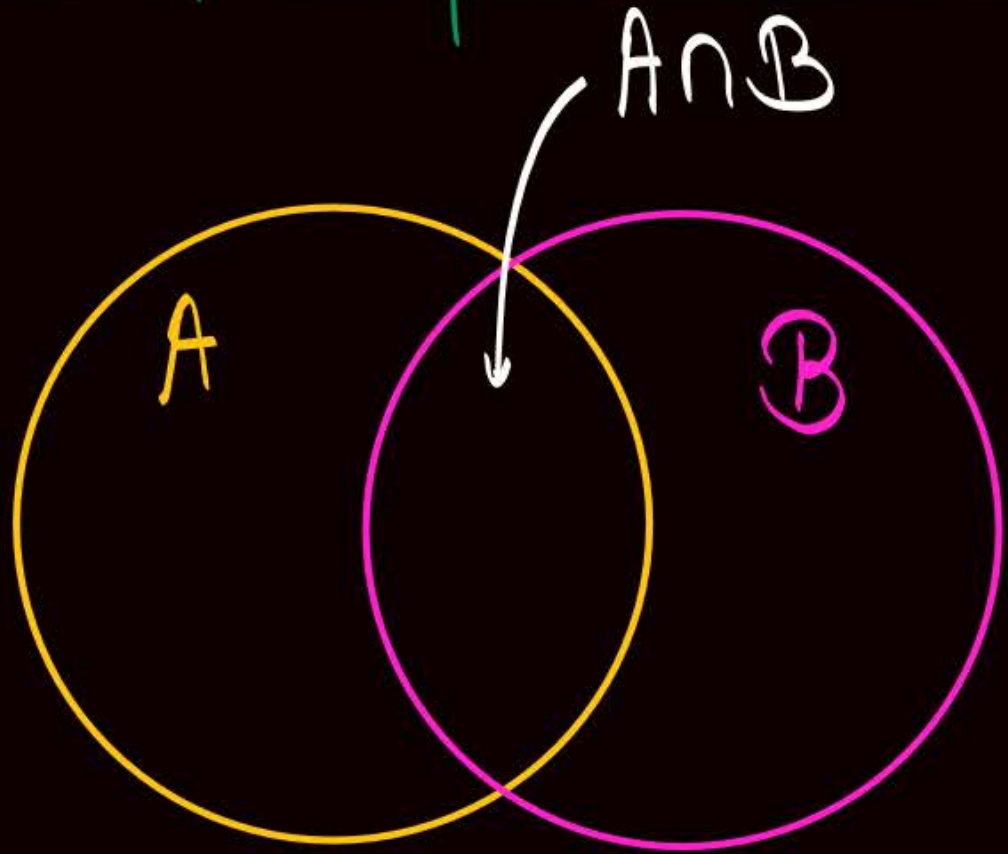
eg:- $A = \{1, 2\}$, $B = \{a, b, c\}$
 $R = \{(1, a), (2, b), (2, c)\}$
 $R^{-1} = \{(a, 1), (b, 2), (c, 2)\}$

Note:-

Let R is a relation on set A ,
then

$$\text{Symmetric Closure of } R = R \cup R^{-1}$$

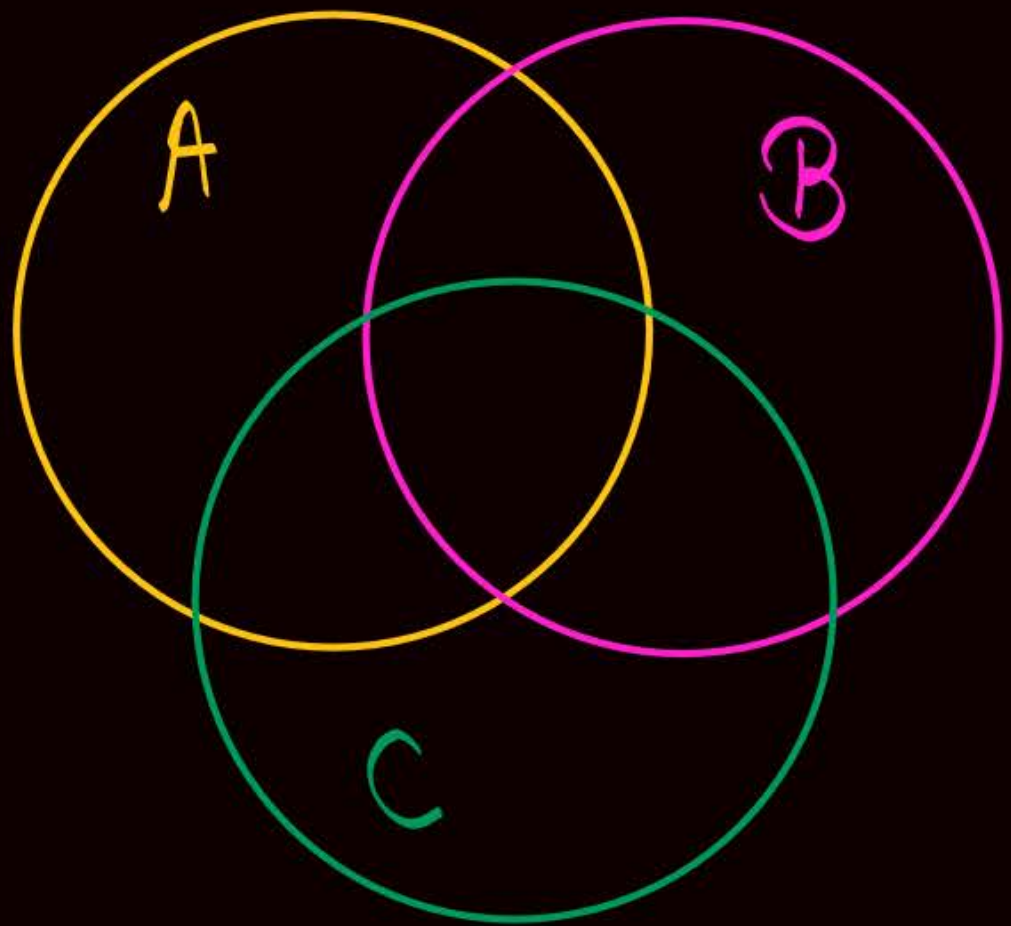
Principle of inclusion & exclusion :-



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Principle of inclusion & Exclusion :-



$$\begin{aligned}n(A \cup B \cup C) &= n(A) + n(B) + n(C) \\&\quad - n(A \cap B) - n(A \cap C) - n(B \cap C) \\&\quad + n(A \cap B \cap C)\end{aligned}$$

14.W.

Q. Let X and Y are two sets such that

$$|X^c| = 15$$

$$|Y^c| = 8$$

and

$$|X \cap Y^c| = 5$$

$$\text{then } |X^c \cap Y| = ?$$

H.W

Q.

The expression

$$(A \cup B \cup C') \cap ((A' \cap B' \cap C) \cup C')$$

is equivalent to.

- (a) $(A \cup B) - C'$
- (b) $(A \cup B) \cap C$
- (c) $(A \cup B) \cap C'$
- (d) None of the above

H.W.

MSQ
Q.:

Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule $(a, b) \in R$ if and only if blood group of a is same as blood group of b

then Relation R is/an

- (a) Reflexive
- (b) Symmetric
- (c) Transitive
- (d) Anti-symmetric

H.W MSQ Q: Let 'P' be the set of all human beings, and R is the relation defined on set 'P' by the rule $(a, b) \in R$ if and only if 'a' is brother of 'b'

then Relation R is

- (a) Reflexive
- (b) Symmetric
- (c) Transitive
- (d) Anti-symmetric
- (e) None of the above



2 mins Summary



✓
Topic

Transitive relation

✓
Topic

Reflexive closure

✓
Topic

Symmetric closure

✓
Topic

Transitive closure

✓
Topic

Complement of a relation and Inverse of a relation

THANK - YOU